Coinbase: AggregateVerifier

Security Review

Cantina Managed review by:
0xIcingdeath, Lead Security Researcher
Rvierdiiev, Security Researcher

April 16, 2026

Contents

1 Introduction

1.1 About Cantina

Cantina is a security services marketplace that connects top security researchers and solutions with clients. Learn more at cantina.xyz

1.2 Disclaimer

Cantina Managed provides a detailed evaluation of the security posture of the code at a particular moment based on the information available at the time of the review. While Cantina Managed endeavors to identify and disclose all potential security issues, it cannot guarantee that every vulnerability will be detected or that the code will be entirely secure against all possible attacks. The assessment is conducted based on the specific commit and version of the code provided. Any subsequent modifications to the code may introduce new vulnerabilities that were absent during the initial review. Therefore, any changes made to the code require a new security review to ensure that the code remains secure. Please be advised that the Cantina Managed security review is not a replacement for continuous security measures such as penetration testing, vulnerability scanning, and regular code reviews.

1.3 Risk assessment

Severity level	**Impact: High**	**Impact: Medium**	**Impact: Low**
Likelihood: high	Critical	High	Medium
Likelihood: medium	High	Medium	Low
Likelihood: low	Medium	Low	Low

1.3.1 Severity Classification

The severity of security issues found during the security review is categorized based on the above table. Critical findings have a high likelihood of being exploited and must be addressed immediately. High findings are almost certain to occur, easy to perform, or not easy but highly incentivized thus must be fixed as soon as possible.

Medium findings are conditionally possible or incentivized but are still relatively likely to occur and should be addressed. Low findings are a rare combination of circumstances to exploit, or offer little to no incentive to exploit but are recommended to be addressed.

Lastly, some findings might represent objective improvements that should be addressed but do not impact the project's overall security (Gas and Informational findings).

2 Security Review Summary

Base is a secure, low-cost, builder-friendly Ethereum L2 built to bring the next billion users onchain.

From Apr 10th to Apr 13th the Cantina team conducted a review of contracts on commit hash ffe2af8c. The team identified a total of **1** issues:

Issues Found

Severity	Count	Fixed	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	0	0	0
Gas Optimizations	0	0	0
Informational	1	1	0
Total	**1**	**1**	**0**

2.1 Scope

The security review had the following components in scope for contracts on commit hash ffe2af8c:

```
src/multiproof/AggregateVerifier.sol
```

3 Findings

3.1 Informational

3.1.1 Document offsets, size, and expected structs

Severity: Informational

Context: AggregateVerifier.sol#L29

Description: The `initializeWithData` function highlights the expected `proof` values, including selector, creator, root claim, extradata, intermediate roots, and cwia bytes. In practice, the contract stores all the above, without the selector, hence all offsets are shifted 4 bytes to the left.

For documentation purposes, we would recommend adding a section in the contract that highlights the data offset, size, and item being stored. This helps with double checking that the getters are retrieving the correct bytes of correct size.

offset (calldata)	size	offset (data)			
0x0	0x04		selector		
0x04	0x14	0x0	creator address		
0x18	0x20	0x14	root claim		
0x38	0x20	0x34	l1 head		
0x58	0x20	0x54	l2 block num	extradata	
0x78	0x14	0x74	parent game	extradata	
0x8c	0x20*number_roots	0x88	intermediate roo	extradata	
		0x20 * x	intermediate roots		
extradata =	0x54	location of l2 block num			confirmed
	0x34	length of l2 block num + parent game			
	0x88	intermediate roots			
gameCreator =	0x0				confirmed
rootClaim =	0x14				confirmed
l1head =	0x34				confirmed
l2SeqNum =	0x54				confirmed
parentAddr =	0x74				confirmed

Recommendation: Consider adding the above as in-code documentation (e.g., in a comment) or in external documentation.

Coinbase: CWIA data offset added as per recommendations in attached PR.

Cantina Managed: Fix verified.